2022-2023 SPRING SEMESTER FLUID MECHANICS MIDTERM1, Date:25/04/2023

Name: **Surname:** **Student ID:**– – – – – – – – –
c b a

If a, b and/or c are zero in your student ID number (xxxxcba), please take it/them as 1.

a; the last digit of your student ID number

b; the penultimate digit of your student ID number

c; the second before the last digit of your student ID number

Question 3 (25p): The x,y, and z components of the velocity field are given by:

$u=ay^2+bx$

$v=bx+c$

$w=0$

a) Is this flow steady or unsteady?
b) Is this flow 1-dimensional, 2-dimensional, or 3-dimensional?
c) Determine the total (material) acceleration in the x-direction.
d) Determine the resultant velocity vector for the location of (x,y,z)=(b, a, 0).
e) Determine the local acceleration vector for the location of (x,y,z)=(b, a, 0).
f) Determine the convective acceleration vector for the location of (x,y,z)=(b, a, 0).
g) Write the equation of the streamline for this flow.